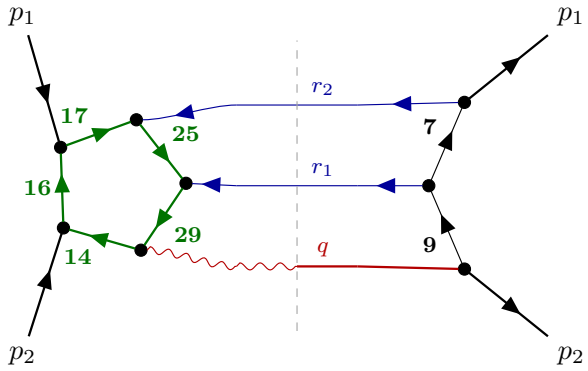
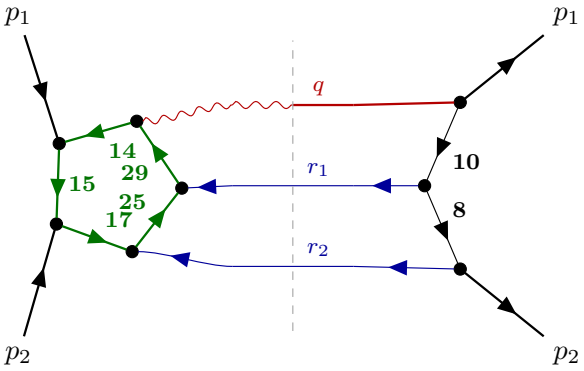
RRV integral families as cut diagrams

One-loop amplitude (left) × tree conjugate (right) for $p_1 p_2 \to V(q)\, r_1 r_2$, glued along the cut (dashed): r_1, r_2 (blue), q (red). The loop lines (green) are the propagators $(k_1 + S)^2$ of the family present in the drawing; the legs entering the loop between two of them meet at one loop vertex, through a tree when an RR propagator of the family separates them. The rapidity condition $-q \cdot p_1 + u\, q \cdot p_2 = 0$ is not a line: the q line is drawn wavy up to the cut ($q^2 = m^2$) and as a double line after it. Arrows: momentum flow with all legs incoming ($p_1 + p_2 + r_1 + r_2 + q = 0$, k_1 incoming); the conjugate is the mirror image. **Propagators** (DD numbers, all legs incoming): $5 = (p_1 + r_1)^2$; $6 = (p_2 + r_1)^2$; $7 = (p_1 + r_2)^2$; $8 = (p_2 + r_2)^2$; $9 = (p_1 + r_1 + r_2)^2$; $10 = (p_2 + r_1 + r_2)^2$; $11 = (r_1 + r_2)^2$; $12 = (p_1 + p_2 + r_1)^2$; $13 = (p_1 + p_2 + r_2)^2$; $14 = (k_1)^2$; $15 = (k_1 + p_1)^2$; $16 = (k_1 + p_2)^2$; $17 = (k_1 + p_1 + p_2)^2$; $18 = (k_1 + r_1)^2$; $19 = (k_1 + p_1 + r_1)^2$; $20 = (k_1 + p_2 + r_1)^2$; $21 = (k_1 + p_1 + p_2 + r_1)^2$; $22 = (k_1 + r_2)^2$; $23 = (k_1 + p_1 + r_2)^2$; $24 = (k_1 + p_2 + r_2)^2$; $25 = (k_1 + p_1 + p_2 + r_2)^2$; $26 = (k_1 + r_1 + r_2)^2$; $27 = (k_1 + p_1 + r_1 + r_2)^2$; $28 = (k_1 + p_2 + r_1 + r_2)^2$; $29 = (k_1 + p_1 + p_2 + r_1 + r_2)^2$.



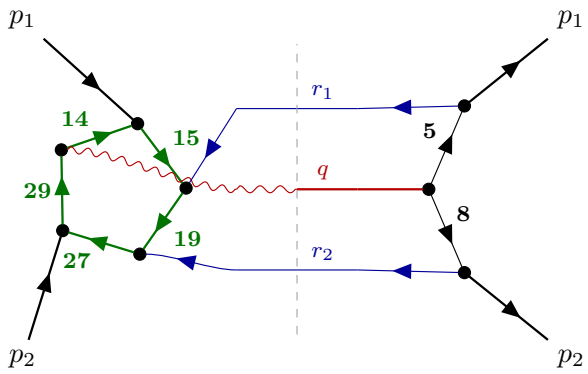
topo1 (RR family 33, pentagon q-p2-p1-r2-r1)

14	$(k_1)^2$	loop
16	$(k_1 + p_2)^2$	loop
17	$(k_1 + p_1 + p_2)^2$	loop
25	$(k_1 + p_1 + p_2 + r_2)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
7	$(p_1 + r_2)^2$	conjugate
9	$(p_1 + r_1 + r_2)^2$	conjugate
13	$(p_1 + p_2 + r_2)^2$	irreducible numerator



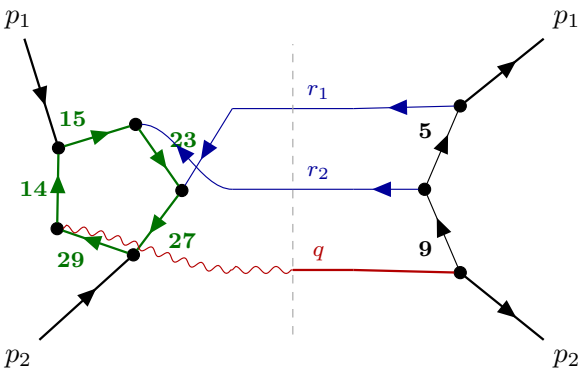
topo2 (RR family 29, pentagon q-p1-p2-r2-r1)

14	$(k_1)^2$	loop
15	$(k_1 + p_1)^2$	loop
17	$(k_1 + p_1 + p_2)^2$	loop
25	$(k_1 + p_1 + p_2 + r_2)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
8	$(p_2 + r_2)^2$	conjugate
10	$(p_2 + r_1 + r_2)^2$	conjugate
13	$(p_1 + p_2 + r_2)^2$	irreducible numerator



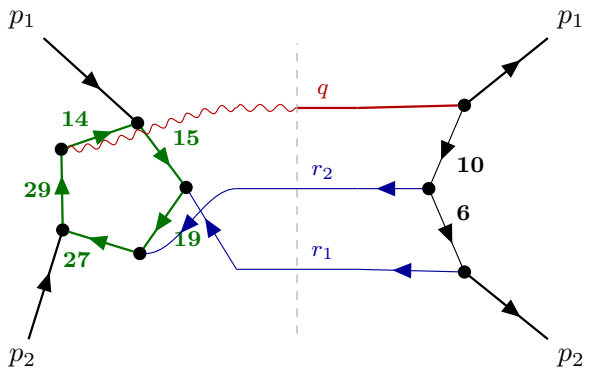
topo3 (RR family 8, pentagon q-p1-r1-r2-p2)

14	$(k_1)^2$	loop
15	$(k_1 + p_1)^2$	loop
19	$(k_1 + p_1 + r_1)^2$	loop
27	$(k_1 + p_1 + r_1 + r_2)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
5	$(p_1 + r_1)^2$	conjugate
8	$(p_2 + r_2)^2$	conjugate
11	$(r_1 + r_2)^2$	irreducible numerator



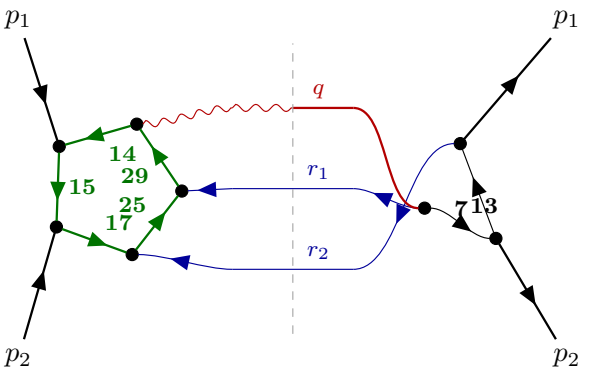
topo4 (RR family 3, pentagon q-p1-r2-r1-p2)

14	$(k_1)^2$	loop
15	$(k_1 + p_1)^2$	loop
23	$(k_1 + p_1 + r_2)^2$	loop
27	$(k_1 + p_1 + r_1 + r_2)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
5	$(p_1 + r_1)^2$	conjugate
9	$(p_1 + r_1 + r_2)^2$	conjugate
6	$(p_2 + r_1)^2$	irreducible numerator



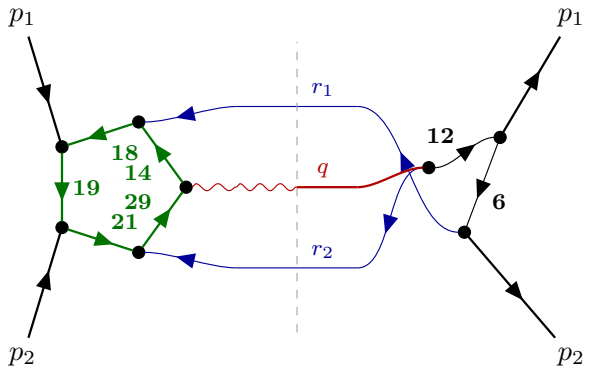
topo5 (RR family 4, pentagon q-p1-r1-r2-p2)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate
5	$(p_1+r_1)^2$	irreducible numerator



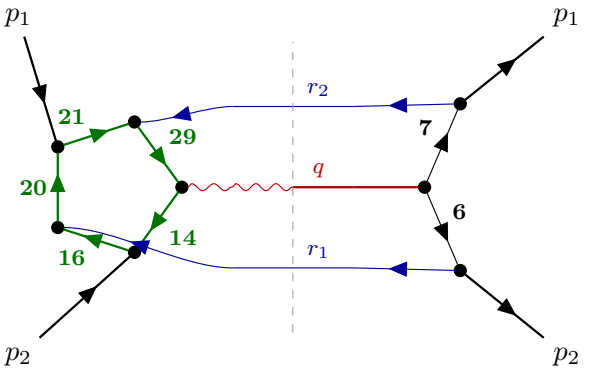
topo6 (RR family 30, pentagon q-p1-p2-r2-r1)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate
11	$(r_1+r_2)^2$	irreducible numerator



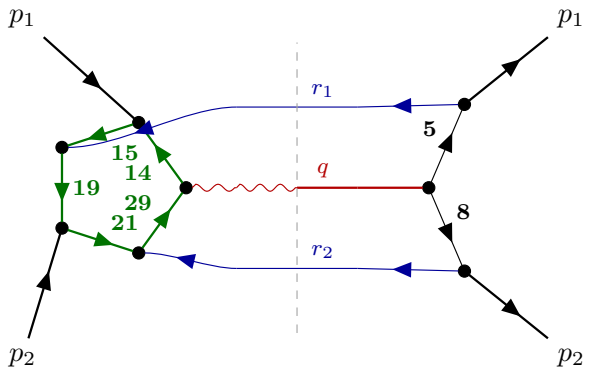
topo7 (RR family 24, pentagon q-r1-p1-p2-r2)

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	irreducible numerator



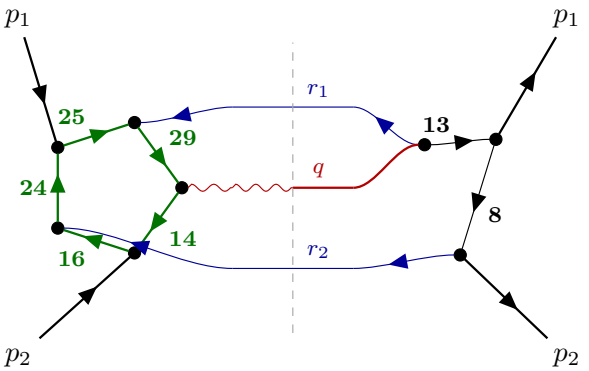
topo8 (RR family 1, pentagon q-p2-r1-p1-r2)

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
7	$(p_1+r_2)^2$	conjugate
5	$(p_1+r_1)^2$	irreducible numerator



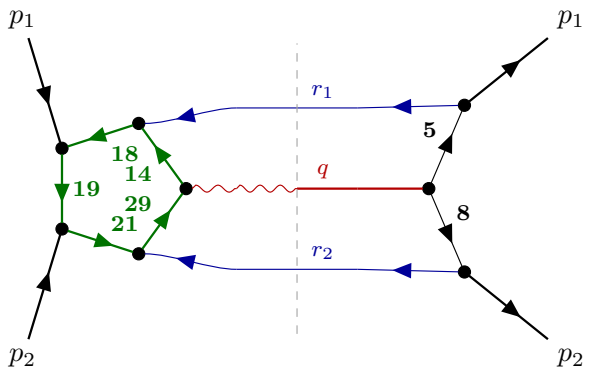
topo9 (RR family 7, pentagon q-p1-r1-p2-r2)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	irreducible numerator



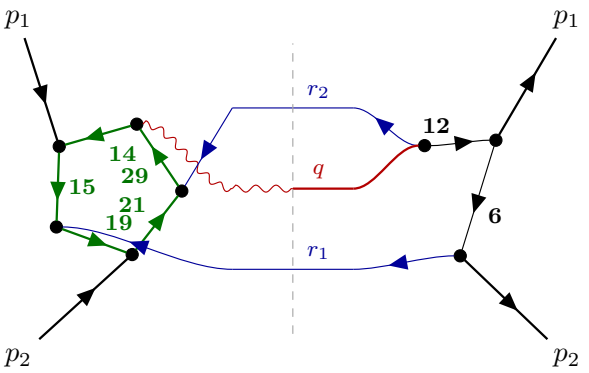
topo10 (RR family 34, pentagon q-p2-r2-p1-r1)

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	irreducible numerator



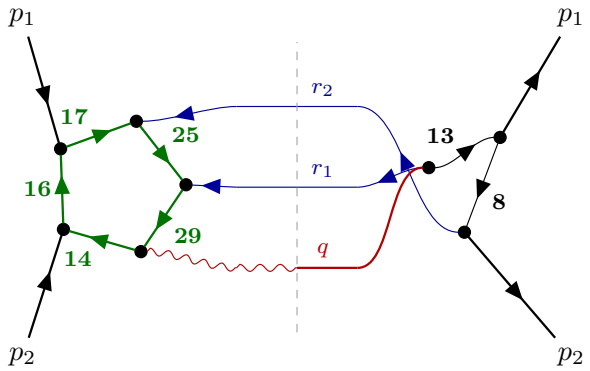
topo11 (RR family 17, pentagon q-r1-p1-p2-r2)

14	$(k_1)^2$	loop
18	$(k_1 + r_1)^2$	loop
19	$(k_1 + p_1 + r_1)^2$	loop
21	$(k_1 + p_1 + p_2 + r_1)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
5	$(p_1 + r_1)^2$	conjugate
8	$(p_2 + r_2)^2$	conjugate
12	$(p_1 + p_2 + r_1)^2$	irreducible numerator



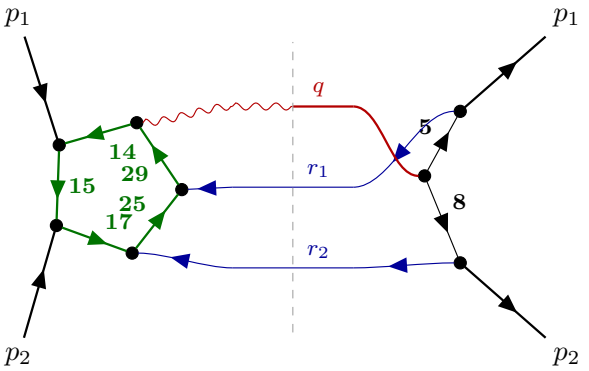
topo12 (RR family 21, pentagon q-p1-r1-p2-r2)

14	$(k_1)^2$	loop
15	$(k_1 + p_1)^2$	loop
19	$(k_1 + p_1 + r_1)^2$	loop
21	$(k_1 + p_1 + p_2 + r_1)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
6	$(p_2 + r_1)^2$	conjugate
12	$(p_1 + p_2 + r_1)^2$	conjugate
8	$(p_2 + r_2)^2$	irreducible numerator



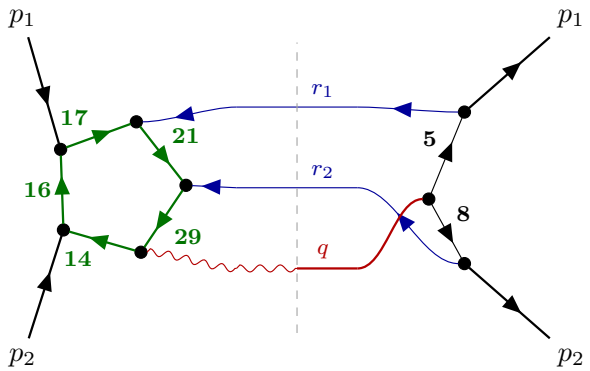
topo13 (RR family 31, pentagon q-p2-p1-r2-r1)

14	$(k_1)^2$	loop
16	$(k_1 + p_2)^2$	loop
17	$(k_1 + p_1 + p_2)^2$	loop
25	$(k_1 + p_1 + p_2 + r_2)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
8	$(p_2 + r_2)^2$	conjugate
13	$(p_1 + p_2 + r_2)^2$	conjugate
11	$(r_1 + r_2)^2$	irreducible numerator



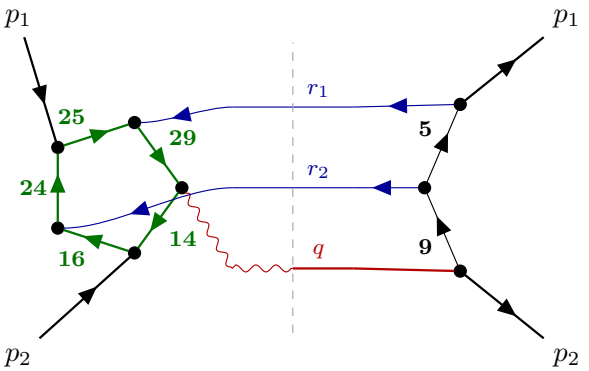
topo14 (RR family 6, pentagon q-p1-p2-r2-r1)

14	$(k_1)^2$	loop
15	$(k_1 + p_1)^2$	loop
17	$(k_1 + p_1 + p_2)^2$	loop
25	$(k_1 + p_1 + p_2 + r_2)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
5	$(p_1 + r_1)^2$	conjugate
8	$(p_2 + r_2)^2$	conjugate
9	$(p_1 + r_1 + r_2)^2$	irreducible numerator



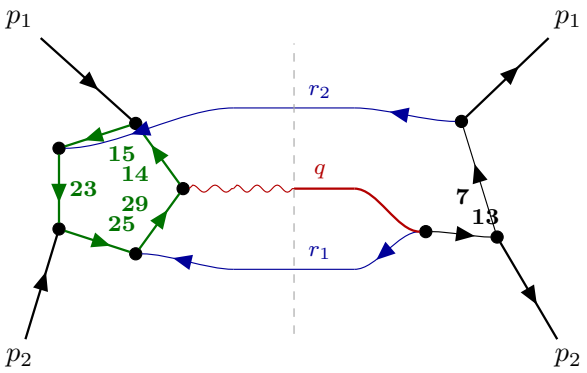
topo15 (RR family 18, pentagon q-p2-p1-r1-r2)

14	$(k_1)^2$	loop
16	$(k_1 + p_2)^2$	loop
17	$(k_1 + p_1 + p_2)^2$	loop
21	$(k_1 + p_1 + p_2 + r_1)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
5	$(p_1 + r_1)^2$	conjugate
8	$(p_2 + r_2)^2$	conjugate
13	$(p_1 + p_2 + r_2)^2$	irreducible numerator



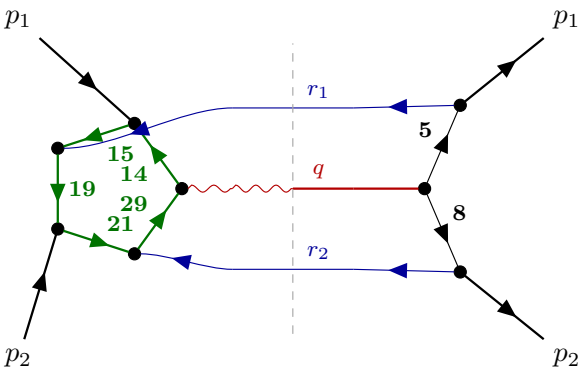
topo16 (RR family 32, pentagon q-p2-r2-p1-r1)

14	$(k_1)^2$	loop
16	$(k_1 + p_2)^2$	loop
24	$(k_1 + p_2 + r_2)^2$	loop
25	$(k_1 + p_1 + p_2 + r_2)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
5	$(p_1 + r_1)^2$	conjugate
9	$(p_1 + r_1 + r_2)^2$	conjugate
13	$(p_1 + p_2 + r_2)^2$	irreducible numerator



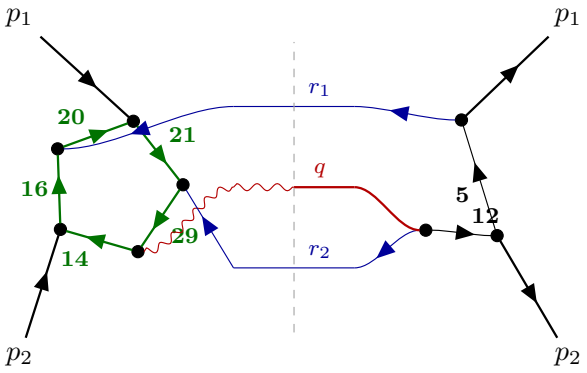
topo17 (RR family 28, pentagon q-p1-r2-p2-r1)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	irreducible numerator



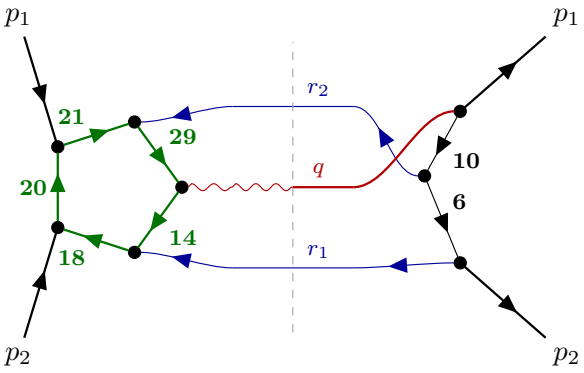
topo18 (RR family 2, pentagon q-p1-r1-p2-r2)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate
6	$(p_2+r_1)^2$	irreducible numerator



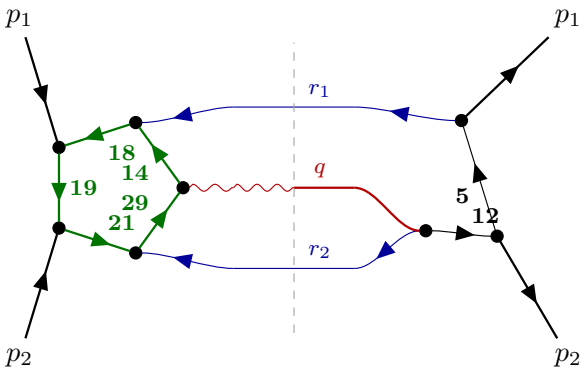
topo19 (RR family 15, pentagon q-p2-r1-p1-r2)

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate
7	$(p_1+r_2)^2$	irreducible numerator



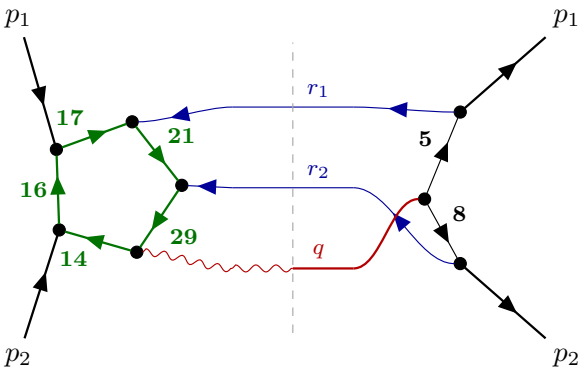
topo20 (RR family 27, pentagon q-r1-p2-p1-r2)

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	irreducible numerator



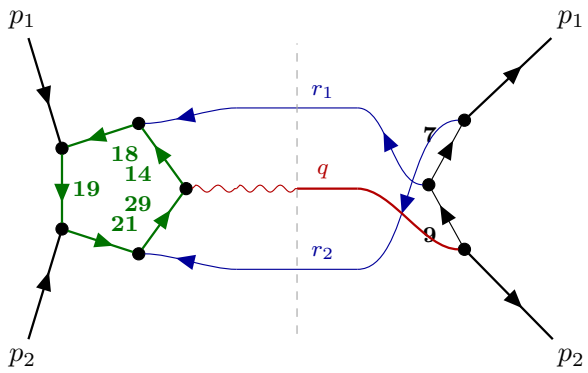
topo21 (RR family 23, pentagon q-r1-p1-p2-r2)

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	irreducible numerator



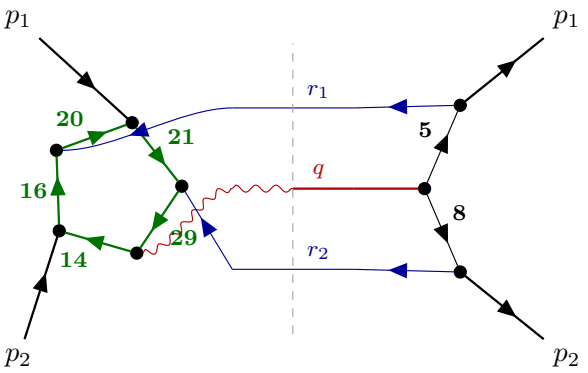
topo22 (RR family 7, pentagon q-p2-p1-r1-r2)

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	irreducible numerator



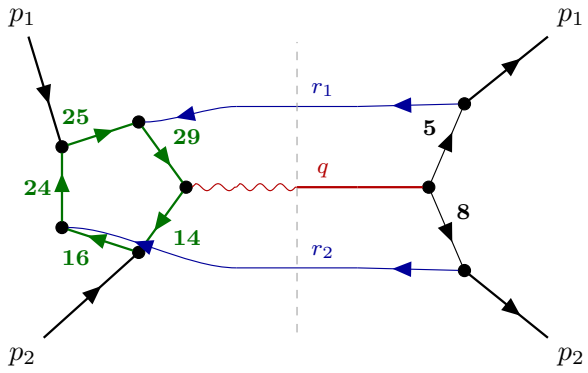
topo23 (RR family 33, pentagon q-r1-p1-p2-r2)

14	$(k_1)^2$	loop
18	$(k_1 + r_1)^2$	loop
19	$(k_1 + p_1 + r_1)^2$	loop
21	$(k_1 + p_1 + p_2 + r_1)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
7	$(p_1 + r_2)^2$	conjugate
9	$(p_1 + r_1 + r_2)^2$	conjugate
13	$(p_1 + p_2 + r_2)^2$	irreducible numerator



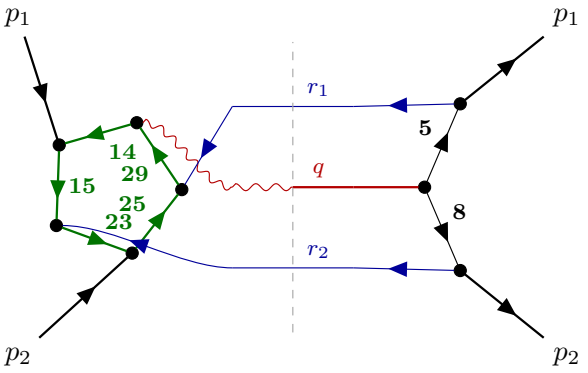
topo24 (RR family 6, pentagon q-p2-r1-p1-r2)

14	$(k_1)^2$	loop
16	$(k_1 + p_2)^2$	loop
20	$(k_1 + p_2 + r_1)^2$	loop
21	$(k_1 + p_1 + p_2 + r_1)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
5	$(p_1 + r_1)^2$	conjugate
8	$(p_2 + r_2)^2$	conjugate
9	$(p_1 + r_1 + r_2)^2$	irreducible numerator



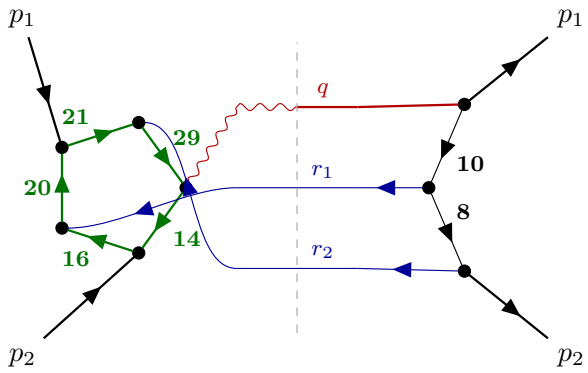
topo25 (RR family 18, pentagon q-p2-r2-p1-r1)

14	$(k_1)^2$	loop
16	$(k_1 + p_2)^2$	loop
24	$(k_1 + p_2 + r_2)^2$	loop
25	$(k_1 + p_1 + p_2 + r_2)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
5	$(p_1 + r_1)^2$	conjugate
8	$(p_2 + r_2)^2$	conjugate
13	$(p_1 + p_2 + r_2)^2$	irreducible numerator



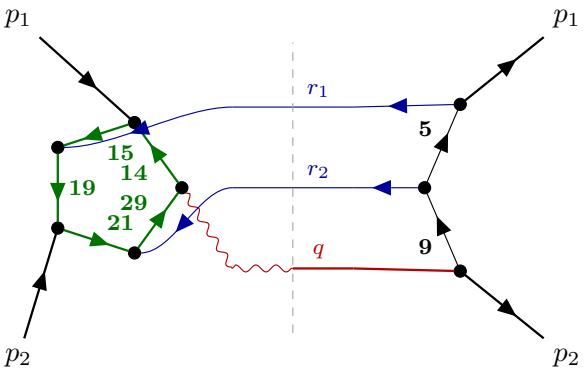
topo26 (RR family 17, pentagon q-p1-r2-p2-r1)

14	$(k_1)^2$	loop
15	$(k_1 + p_1)^2$	loop
23	$(k_1 + p_1 + r_2)^2$	loop
25	$(k_1 + p_1 + p_2 + r_2)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
5	$(p_1 + r_1)^2$	conjugate
8	$(p_2 + r_2)^2$	conjugate
12	$(p_1 + p_2 + r_1)^2$	irreducible numerator



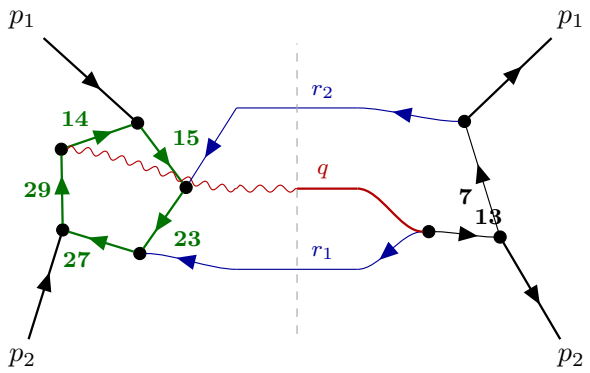
topo27 (RR family 29, pentagon q-p2-r1-p1-r2)

14	$(k_1)^2$	loop
16	$(k_1 + p_2)^2$	loop
20	$(k_1 + p_2 + r_1)^2$	loop
21	$(k_1 + p_1 + p_2 + r_1)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
8	$(p_2 + r_2)^2$	conjugate
10	$(p_2 + r_1 + r_2)^2$	conjugate
13	$(p_1 + p_2 + r_2)^2$	irreducible numerator



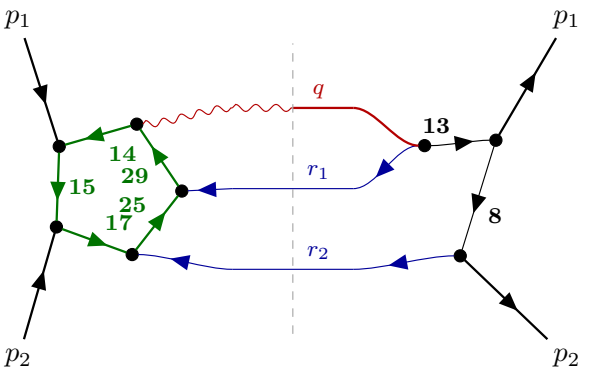
topo28 (RR family 3, pentagon q-p1-r1-p2-r2)

14	$(k_1)^2$	loop
15	$(k_1 + p_1)^2$	loop
19	$(k_1 + p_1 + r_1)^2$	loop
21	$(k_1 + p_1 + p_2 + r_1)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
5	$(p_1 + r_1)^2$	conjugate
9	$(p_1 + r_1 + r_2)^2$	conjugate
6	$(p_2 + r_1)^2$	irreducible numerator



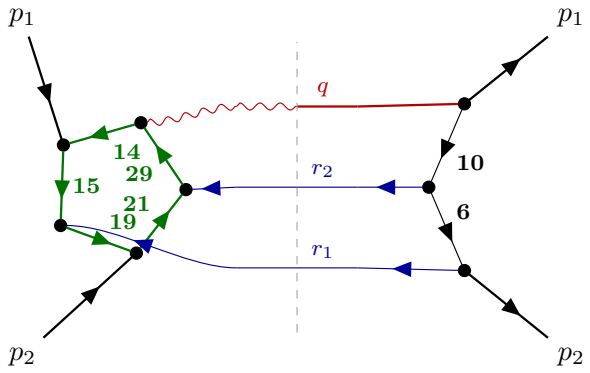
topo29 (RR family 30, pentagon q-p1-r2-r1-p2)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate
11	$(r_1+r_2)^2$	irreducible numerator



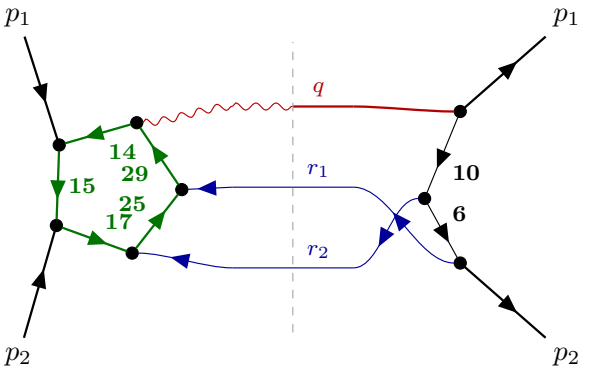
topo30 (RR family 31, pentagon q-p1-p2-r2-r1)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate
11	$(r_1+r_2)^2$	irreducible numerator



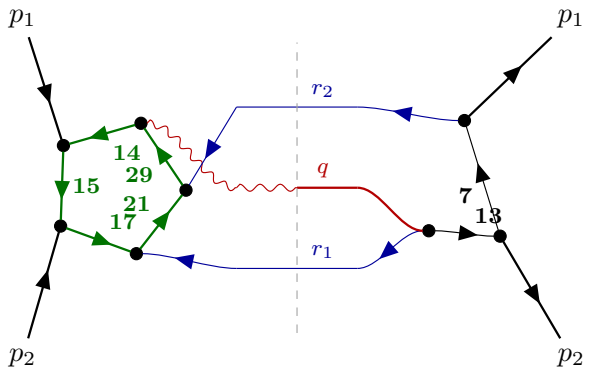
topo31 (RR family 4, pentagon q-p1-r1-p2-r2)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate
5	$(p_1+r_1)^2$	irreducible numerator



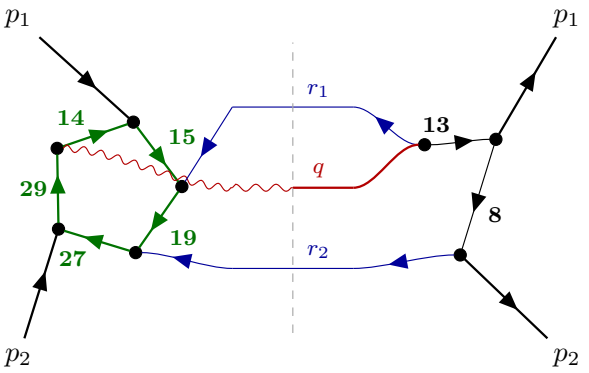
topo32 (RR family 27, pentagon q-p1-p2-r2-r1)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	irreducible numerator



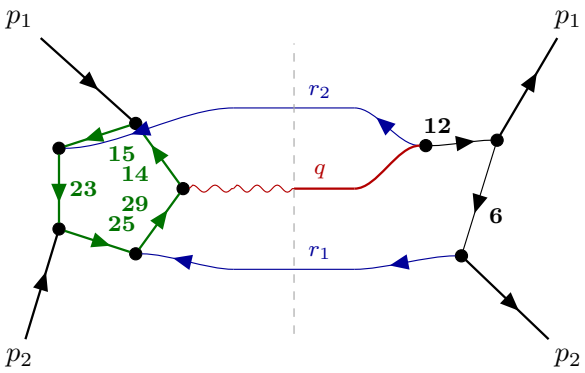
topo33 (RR family 28, pentagon q-p1-p2-r1-r2)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	irreducible numerator



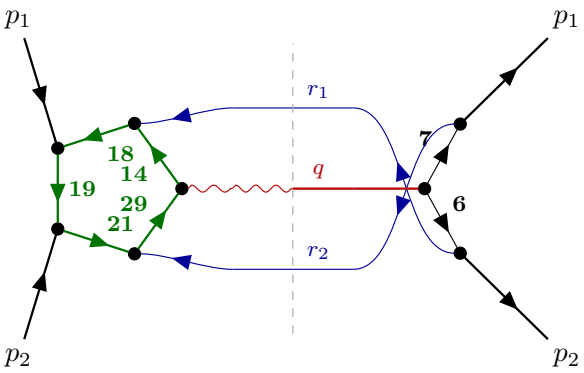
topo34 (RR family 34, pentagon q-p1-r1-r2-p2)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	irreducible numerator



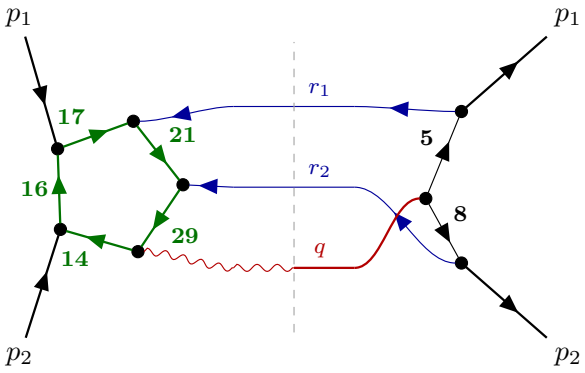
topo35 (RR family 24, pentagon q-p1-r2-p2-r1)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	irreducible numerator



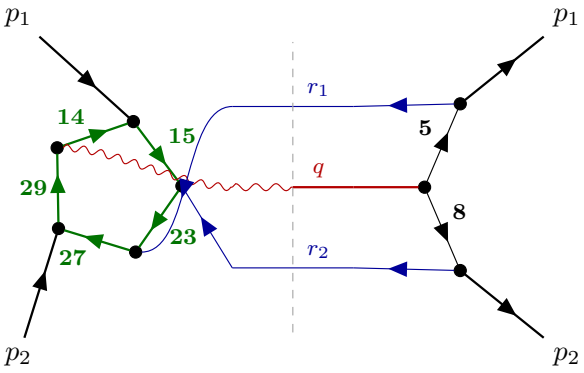
topo36 (RR family 1, pentagon q-r1-p1-p2-r2)

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
7	$(p_1+r_2)^2$	conjugate
5	$(p_1+r_1)^2$	irreducible numerator



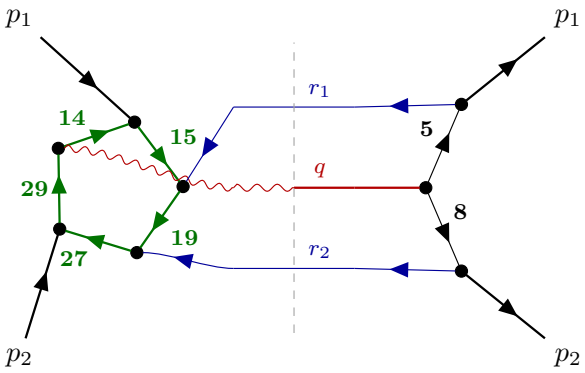
topo37 (RR family 8, pentagon q-p2-p1-r1-r2)

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate
11	$(r_1+r_2)^2$	irreducible numerator



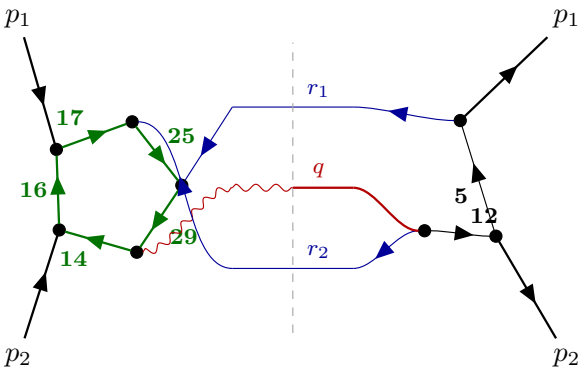
topo38 (RR family 2, pentagon q-p1-r2-r1-p2)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate
6	$(p_2+r_1)^2$	irreducible numerator



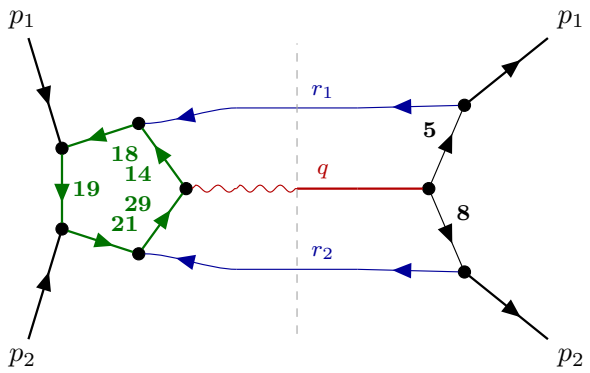
topo39 (RR family 6, pentagon q-p1-r1-r2-p2)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	irreducible numerator



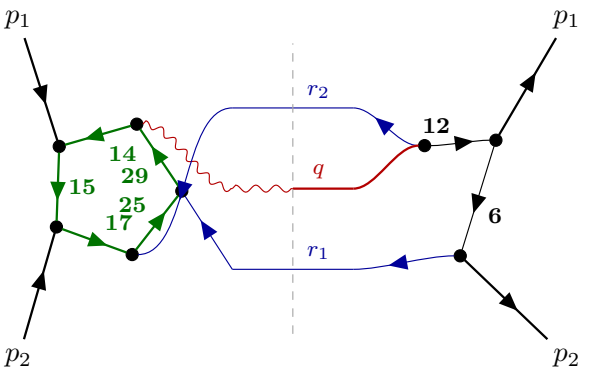
topo40 (RR family 15, pentagon q-p2-p1-r2-r1)

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate
7	$(p_1+r_2)^2$	irreducible numerator



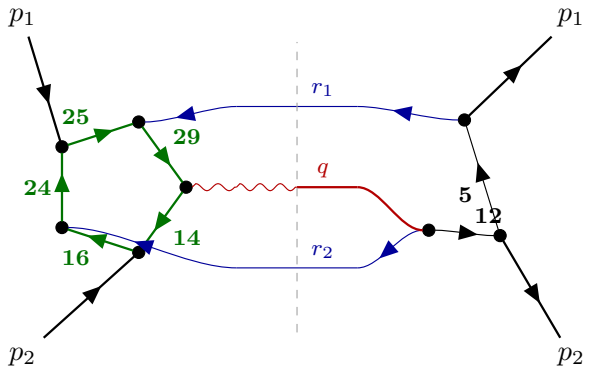
topo41 (RR family 18, pentagon q-r1-p1-p2-r2)

14	$(k_1)^2$	loop
18	$(k_1 + r_1)^2$	loop
19	$(k_1 + p_1 + r_1)^2$	loop
21	$(k_1 + p_1 + p_2 + r_1)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
5	$(p_1 + r_1)^2$	conjugate
8	$(p_2 + r_2)^2$	conjugate
13	$(p_1 + p_2 + r_2)^2$	irreducible numerator



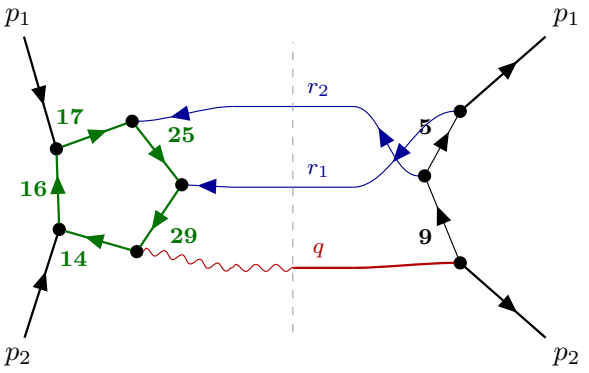
topo42 (RR family 21, pentagon q-p1-p2-r2-r1)

14	$(k_1)^2$	loop
15	$(k_1 + p_1)^2$	loop
17	$(k_1 + p_1 + p_2)^2$	loop
25	$(k_1 + p_1 + p_2 + r_2)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
6	$(p_2 + r_1)^2$	conjugate
12	$(p_1 + p_2 + r_1)^2$	conjugate
8	$(p_2 + r_2)^2$	irreducible numerator



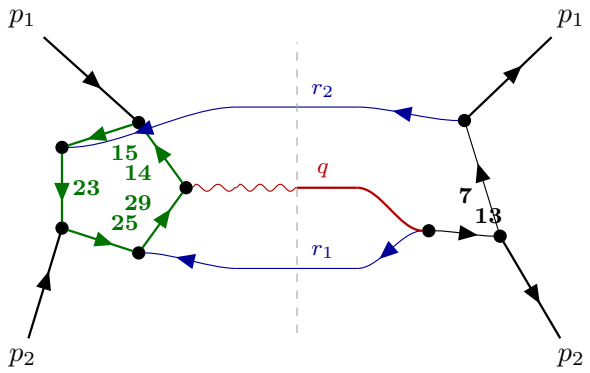
topo43 (RR family 23, pentagon q-p2-r2-p1-r1)

14	$(k_1)^2$	loop
16	$(k_1 + p_2)^2$	loop
24	$(k_1 + p_2 + r_2)^2$	loop
25	$(k_1 + p_1 + p_2 + r_2)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
5	$(p_1 + r_1)^2$	conjugate
12	$(p_1 + p_2 + r_1)^2$	conjugate
13	$(p_1 + p_2 + r_2)^2$	irreducible numerator



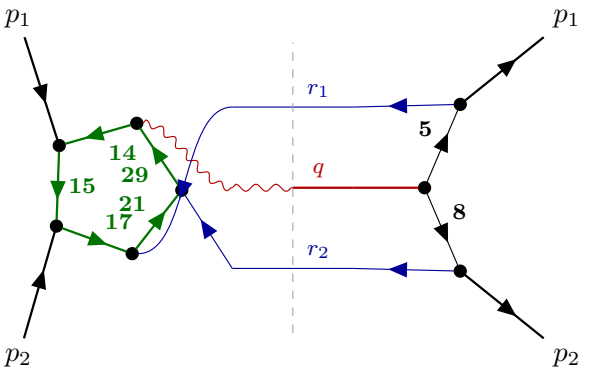
topo44 (RR family 32, pentagon q-p2-p1-r2-r1)

14	$(k_1)^2$	loop
16	$(k_1 + p_2)^2$	loop
17	$(k_1 + p_1 + p_2)^2$	loop
25	$(k_1 + p_1 + p_2 + r_2)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
5	$(p_1 + r_1)^2$	conjugate
9	$(p_1 + r_1 + r_2)^2$	conjugate
13	$(p_1 + p_2 + r_2)^2$	irreducible numerator



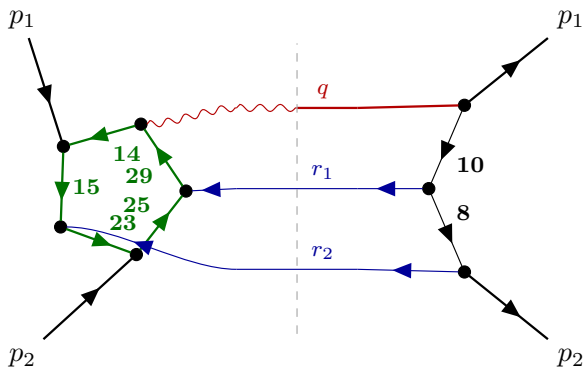
topo45 (RR family 30, pentagon q-p1-r2-p2-r1)

14	$(k_1)^2$	loop
15	$(k_1 + p_1)^2$	loop
23	$(k_1 + p_1 + r_2)^2$	loop
25	$(k_1 + p_1 + p_2 + r_2)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
7	$(p_1 + r_2)^2$	conjugate
13	$(p_1 + p_2 + r_2)^2$	conjugate
11	$(r_1 + r_2)^2$	irreducible numerator



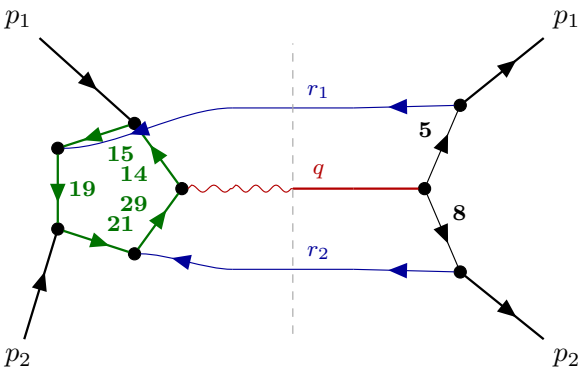
topo46 (RR family 7, pentagon q-p1-p2-r1-r2)

14	$(k_1)^2$	loop
15	$(k_1 + p_1)^2$	loop
17	$(k_1 + p_1 + p_2)^2$	loop
21	$(k_1 + p_1 + p_2 + r_1)^2$	loop
29	$(k_1 + p_1 + p_2 + r_1 + r_2)^2$	loop
5	$(p_1 + r_1)^2$	conjugate
8	$(p_2 + r_2)^2$	conjugate
10	$(p_2 + r_1 + r_2)^2$	irreducible numerator



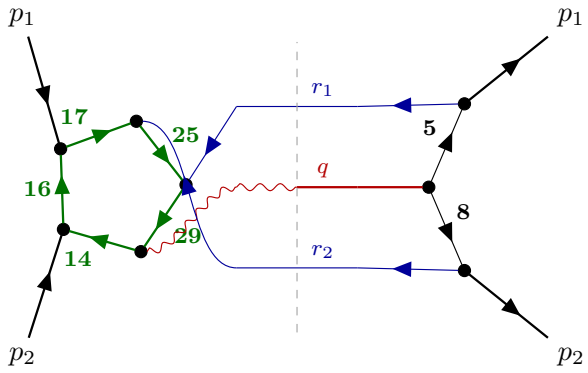
topo47 (RR family 29, pentagon q-p1-r2-p2-r1)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	irreducible numerator



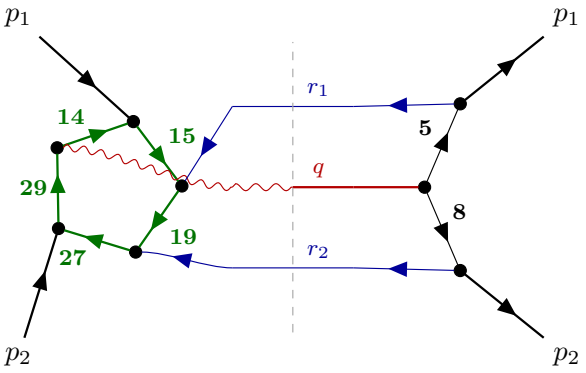
topo48 (RR family 17, pentagon q-p1-r1-p2-r2)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	irreducible numerator



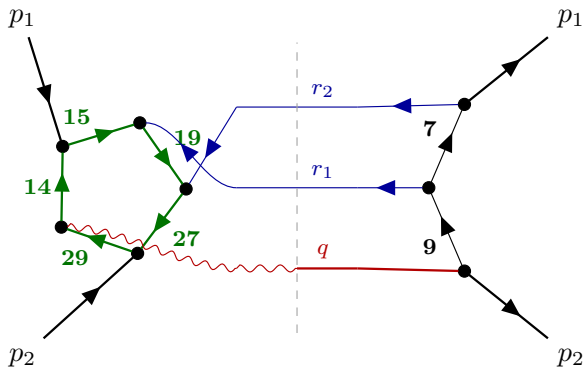
topo49 (RR family 6, pentagon q-p2-p1-r2-r1)

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	irreducible numerator



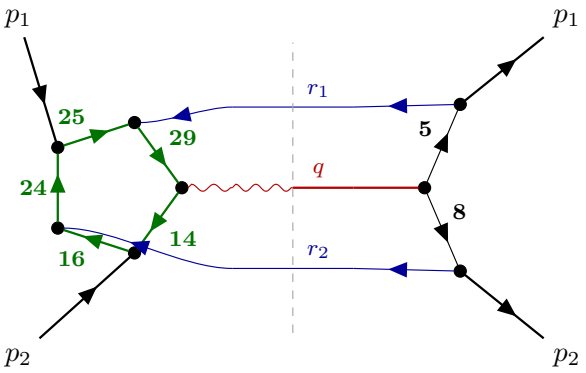
topo50 (RR family 7, pentagon q-p1-r1-r2-p2)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	irreducible numerator



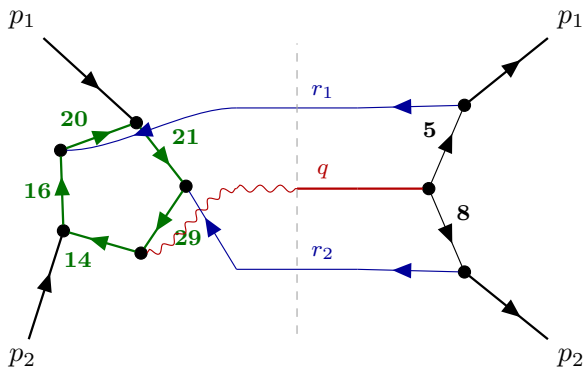
topo51 (RR family 33, pentagon q-p1-r1-r2-p2)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	irreducible numerator



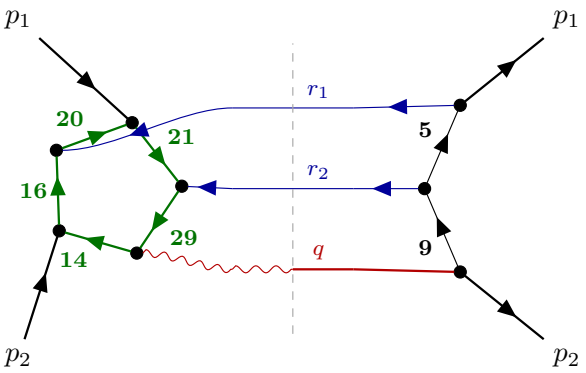
topo52 (RR family 8, pentagon q-p2-r2-p1-r1)

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate
11	$(r_1+r_2)^2$	irreducible numerator



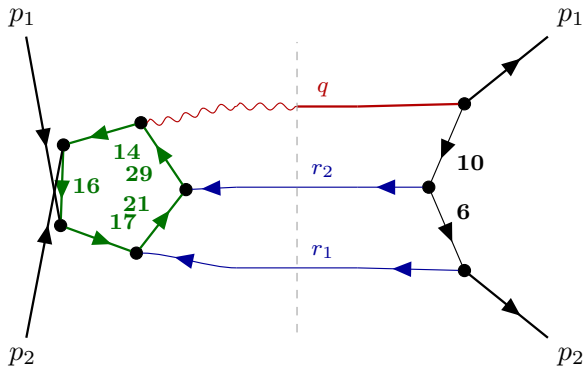
topo53 (RR family 2, pentagon q-p2-r1-p1-r2)

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate
6	$(p_2+r_1)^2$	irreducible numerator



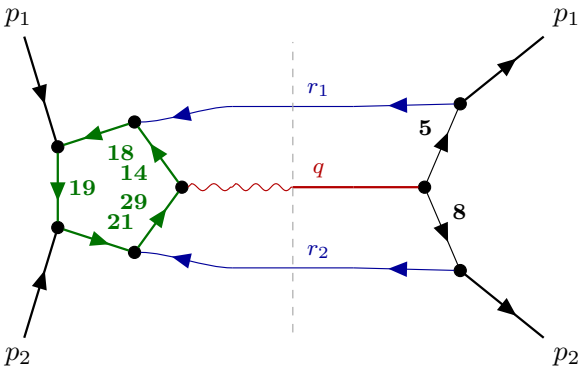
topo54 (RR family 3, pentagon q-p2-r1-p1-r2)

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate
6	$(p_2+r_1)^2$	irreducible numerator



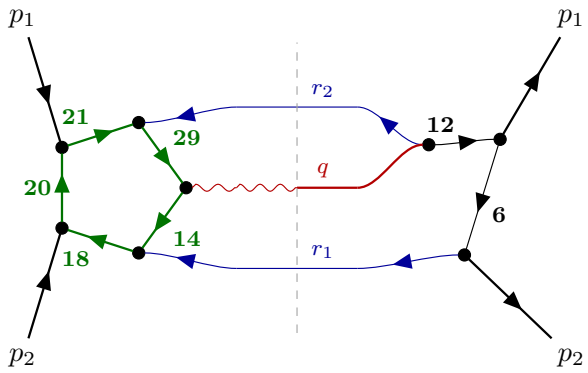
topo55 (RR family 4, pentagon q-p2-p1-r1-r2)

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate
5	$(p_1+r_1)^2$	irreducible numerator



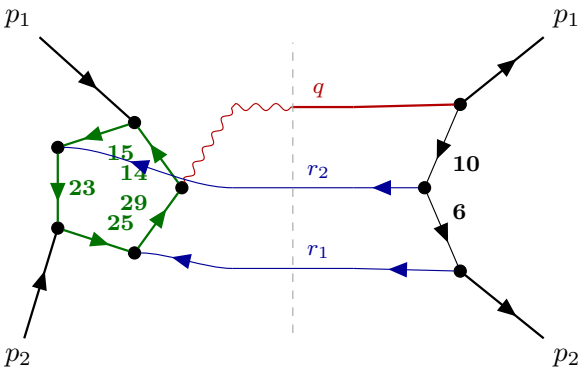
topo56 (RR family 6, pentagon q-r1-p1-p2-r2)

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	irreducible numerator



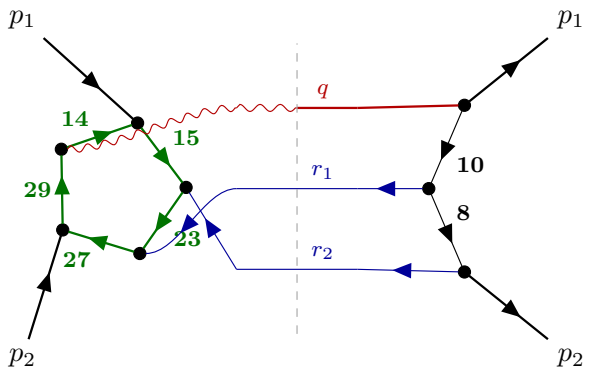
topo57 (RR family 24, pentagon q-r1-p2-p1-r2)

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	irreducible numerator



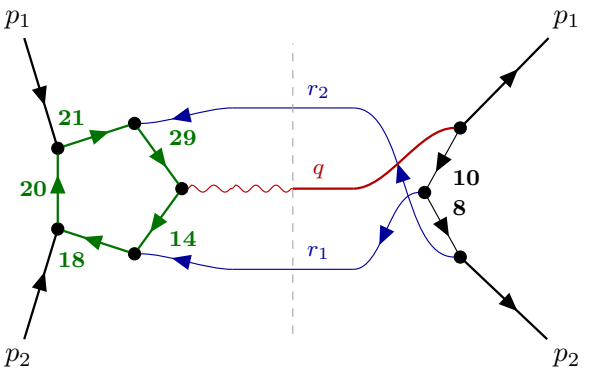
topo58 (RR family 27, pentagon q-p1-r2-p2-r1)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	irreducible numerator



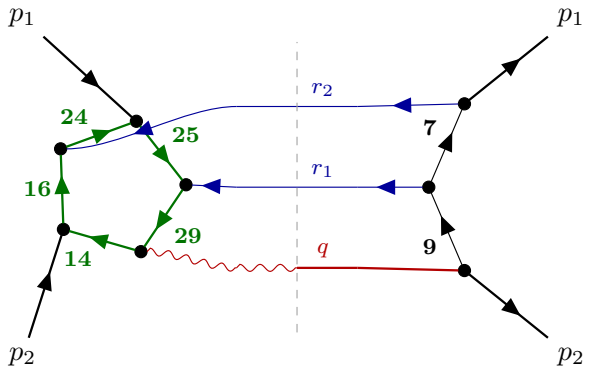
topo59 (RR family 29, pentagon q-p1-r2-r1-p2)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	irreducible numerator



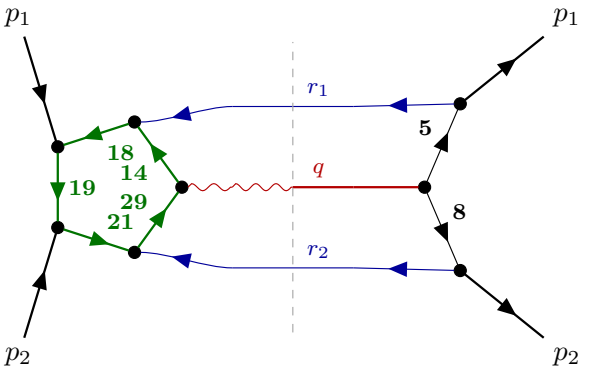
topo60 (RR family 29, pentagon q-r1-p2-p1-r2)

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	irreducible numerator



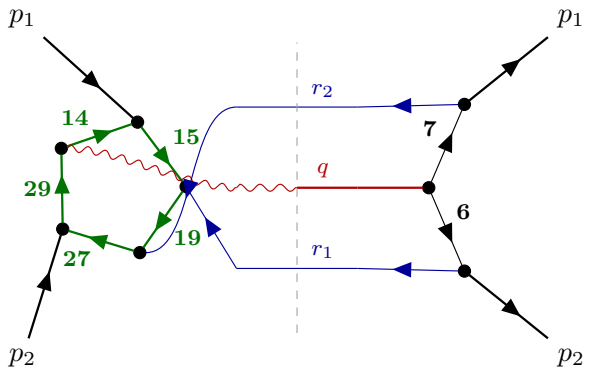
topo61 (RR family 33, pentagon q-p2-r2-p1-r1)

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	irreducible numerator



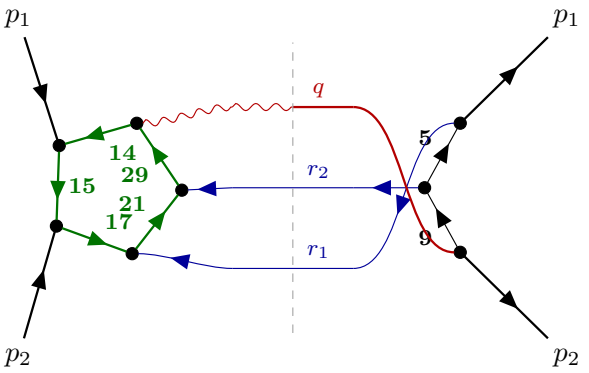
topo62 (RR family 8, pentagon q-r1-p1-p2-r2)

14	$(k_1)^2$	loop
18	$(k_1+r_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate
11	$(r_1+r_2)^2$	irreducible numerator



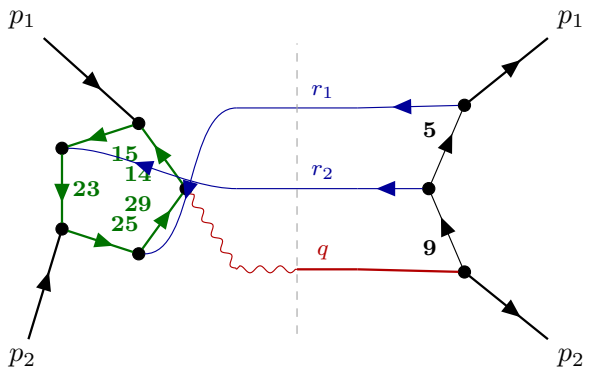
topo63 (RR family 1, pentagon q-p1-r1-r2-p2)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
7	$(p_1+r_2)^2$	conjugate
5	$(p_1+r_1)^2$	irreducible numerator



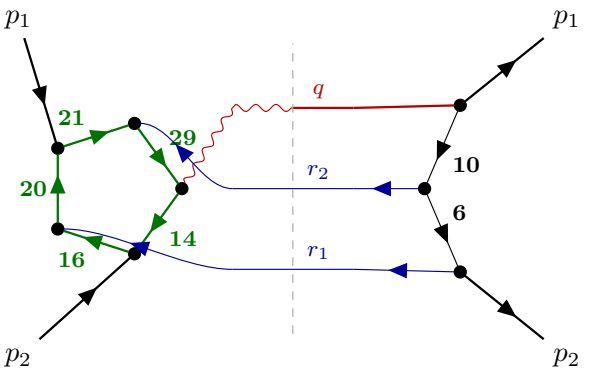
topo64 (RR family 3, pentagon q-p1-p2-r1-r2)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate
6	$(p_2+r_1)^2$	irreducible numerator



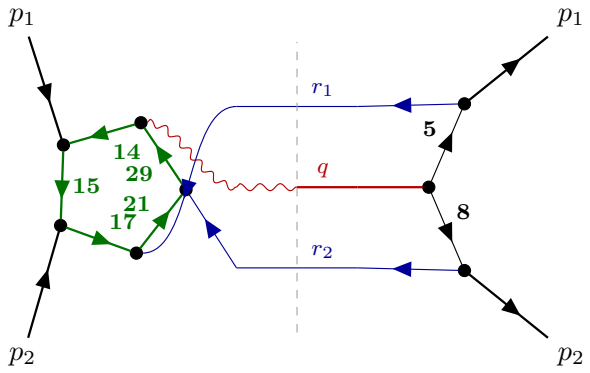
topo65 (RR family 3, pentagon q-p1-r2-p2-r1)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate
6	$(p_2+r_1)^2$	irreducible numerator



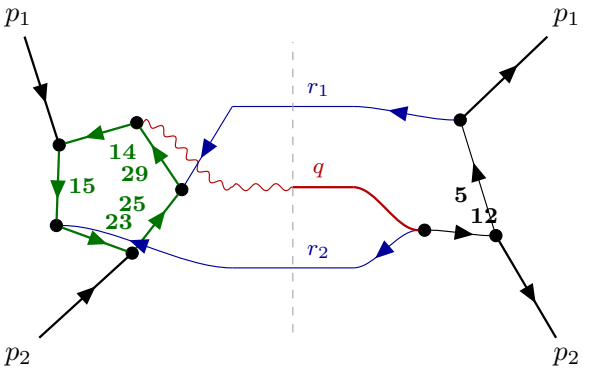
topo66 (RR family 4, pentagon q-p2-r1-p1-r2)

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
20	$(k_1+p_2+r_1)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	conjugate
5	$(p_1+r_1)^2$	irreducible numerator



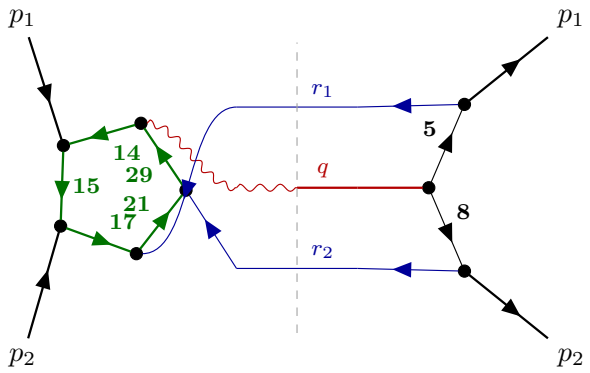
topo67 (RR family 8, pentagon q-p1-p2-r1-r2)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate
11	$(r_1+r_2)^2$	irreducible numerator



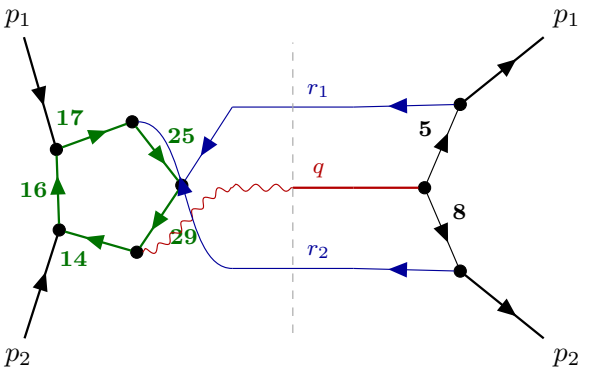
topo68 (RR family 15, pentagon q-p1-r2-p2-r1)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate
7	$(p_1+r_2)^2$	irreducible numerator



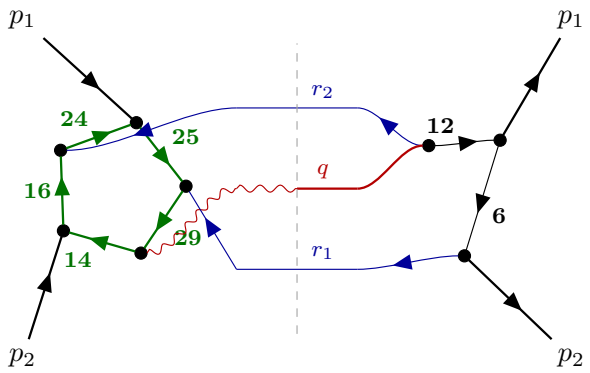
topo69 (RR family 17, pentagon q-p1-p2-r1-r2)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	irreducible numerator



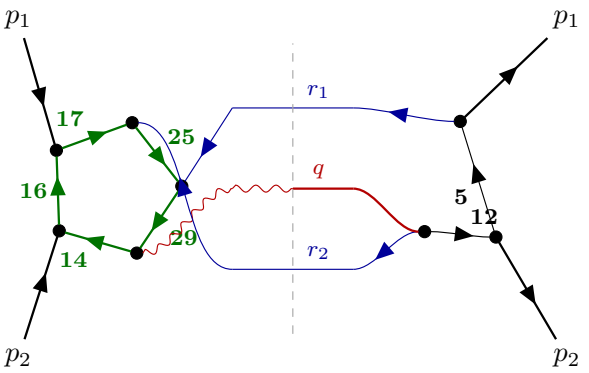
topo70 (RR family 18, pentagon q-p2-p1-r2-r1)

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	irreducible numerator



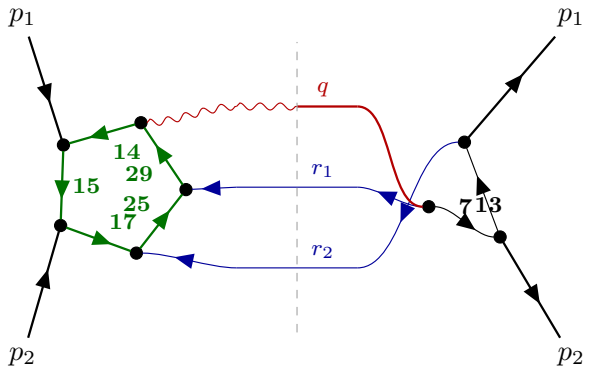
topo71 (RR family 21, pentagon q-p2-r2-p1-r1)

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
24	$(k_1+p_2+r_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
6	$(p_2+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate
8	$(p_2+r_2)^2$	irreducible numerator



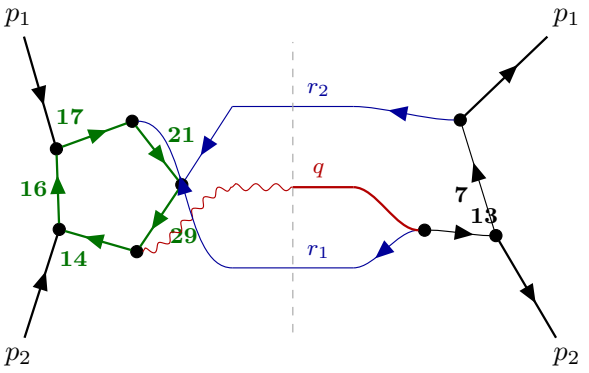
topo72 (RR family 23, pentagon q-p2-p1-r2-r1)

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
12	$(p_1+p_2+r_1)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	irreducible numerator



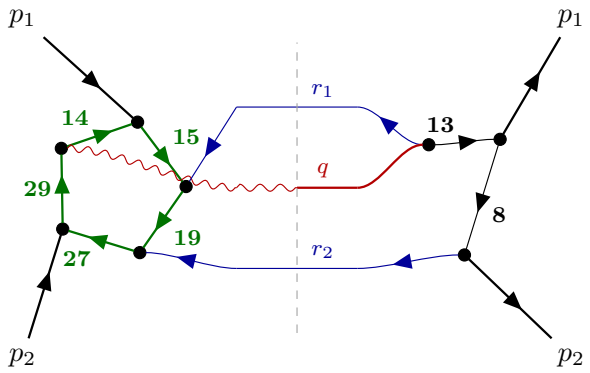
topo73 (RR family 28, pentagon q-p1-p2-r2-r1)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate
10	$(p_2+r_1+r_2)^2$	irreducible numerator



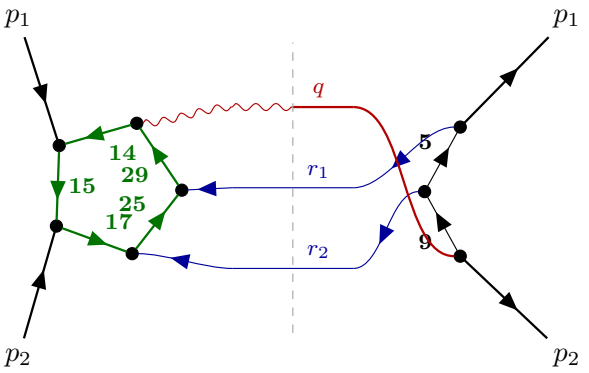
topo74 (RR family 30, pentagon q-p2-p1-r1-r2)

14	$(k_1)^2$	loop
16	$(k_1+p_2)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
21	$(k_1+p_1+p_2+r_1)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
7	$(p_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate
11	$(r_1+r_2)^2$	irreducible numerator



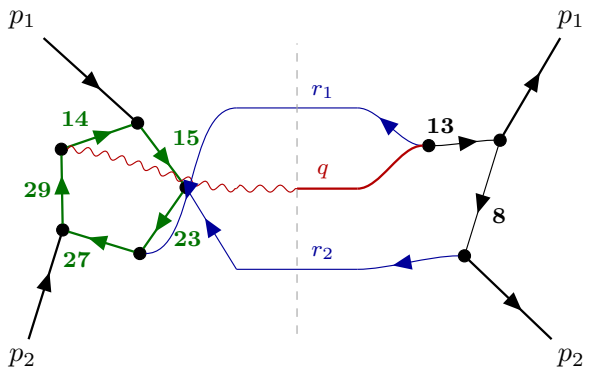
topo75 (RR family 31, pentagon q-p1-r1-r2-p2)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
19	$(k_1+p_1+r_1)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate
11	$(r_1+r_2)^2$	irreducible numerator



topo76 (RR family 32, pentagon q-p1-p2-r2-r1)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
17	$(k_1+p_1+p_2)^2$	loop
25	$(k_1+p_1+p_2+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
5	$(p_1+r_1)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	irreducible numerator



topo77 (RR family 34, pentagon q-p1-r2-r1-p2)

14	$(k_1)^2$	loop
15	$(k_1+p_1)^2$	loop
23	$(k_1+p_1+r_2)^2$	loop
27	$(k_1+p_1+r_1+r_2)^2$	loop
29	$(k_1+p_1+p_2+r_1+r_2)^2$	loop
8	$(p_2+r_2)^2$	conjugate
13	$(p_1+p_2+r_2)^2$	conjugate
9	$(p_1+r_1+r_2)^2$	irreducible numerator